CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of COMPUTER STUDIES

CSAC 212 - Introduction to Artificial Intelligence

Digitized Coin Counting: The Use of Neural Network and Computer Vision in Summing Philippine Peso Coins

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Abstract

The world is moving into a time where technology can be seen everywhere. People use Artificial Intelligence (AI) in their daily lives. The motivation for this project is to make the lives of people with small businesses easier by using AI in coin counting. Digitizing coin counting will save a lot of time and effort compared to manual counting, especially in larger quantities. The study uses a Convolutional Neural Network (CNN) to identify the Philippine peso coins and their corresponding value. The dataset being used is composed of several images of coins varying from 25 cents to 20 peso coins.

1 Introduction

As the world revolutionized towards a more advanced technology, several inventions were created. One of which is Artificial Intelligence (AI). In the present times, AI is evolving to be more human-like in terms of cognitive abilities. Along with this evolution, solutions to world problems are present in our daily lives. An example of this is tallying or counting large quantities of money. It requires a fair amount of time when done manually by humans and time is one of the most important resources in the world, even more important than money. It is important to efficiently calculate or tally your money to save time and do more essential tasks.

The input of the program consists of a labeled data set of several varying images of Philippine Peso coins ranging from five cents to 20 peso coins. The data set can be used in training a convolutional neural network via supervised learning to identify what type of coin is in the image. The processed data will be integrated into the program to calculate the total amount of coins within the image. The expected output is the summed-up amount of coins calculated by the program and the quantities of each coin found by AI. Digitizing the counting of money, especially coins can help many store owners and small businesses that still tally their earnings manually.

2 Related work

This section contains literature and studies related to this study. These related studies are collected from various sources such as Google Scholar, Academia, IEEE Xplore, ScienceDirect, and ResearchGate. These related studies play a crucial role in guiding the researchers to develop the current study further. The following studies provide additional information which helps the reader in understanding this study.

According to Saranya and Subhashini [9], Artificial Intelligence (AI) is commonly used in solving real-world problems. It includes machine learning as well as deep learning topics which are used in predicting outcomes without the help of humans. Their study mainly focuses on Explainable Artificial Intelligence (XAI) models. They discussed many applications of XAI, in agriculture, computer vision, finance, forecasting, healthcare, and more. This study is more focused on different applications of XAI compared to the current study whose main focus is the implementation of AI in camera vision to count peso coins. Another study conducted by Santos et al. [13] presents the use of AI in systems to aid in development of both cognitive and motor skills.

The study of Shreyas et al. [10] revealed that neural network is one of the most common topics in research for its many uses. It covers the topics of medicine, meteorology, image processing, speech recognition, and many different topics. It was also stated that using neural networks gives better output compared to humans in terms of speed, accuracy, and errors. Of all the types of neural networks the convolutional neural network is the one used in image processing and language [5].

Computer vision is mostly used along with AI programs in detecting similarities and differences which then are used in many different real-life problems. Brought by the COVID-19 pandemic are several changes including the mandatory wearing of face masks and social distancing. Eyiokur et al. [4] conducted a study on how to utilize computer vision in face mask detection, face recognition, hand-face interaction, crowd-counting, and pandemic-related problems. Another study presented by Agnew et al. [1] uses computer vision in detecting overfilled trash bins. This study applies computer vision and AI to assist humans in waste management. Both studies utilized computer vision in solving a real-world problem by processing the images and detecting patterns. On the other hand, Waelen [11] stated that relying on computer vision has also its disadvantages, including the lack of privacy.

Prabu et al. [7] conducted a similar study that detects and recognizes Indian coins considering their features such as texture, color, shape, and other characteristics. They used YOLOv5 (a computer vision model that means You Only Look Once version 5) in detecting objects. It was stated that YOLOv5 is comparably faster and more accurate than other models. Another similar study conducted by Rangan [8] utilizes computer vision in sorting out Indian coins. The result was a machine capable of identifying and sorting coins. The only downside was the cost of the creation of the machine. Cooper and Arandjelović [3] had a similar study of identifying coins with the use of computer vision and the convolutional neural network. They used an approach that makes the neural network identify individual parts of the coins to imitate how humans identify a coin.

Studies above show a significant connection with the current study. Artificial intelligence and neural networks are widely used in digitizing work, especially those that are done manually. The use of neural networks along with computer vision in identifying and differentiating an object from its surroundings was shown to be effective.

3 Methods

This section describes the step-by-step process of how neural networks, specifically a convolutional neural network, and computer vision are implemented to train AI that can recognize coins from each other and their surroundings. The trained AI will then be integrated into a program that calculates the total amount of coins detected and identified by AI. In this study, a supervised learning method will be used in training AI to differentiate the coins from other objects and their surroundings.

Convolutional Neural Network (CNN) is an architecture of deep learning that is commonly used in computer vision and natural language [5]. The study will use a machine learning platform called TensorFlow to create and train a convolutional neural network using the supervised learning method to identify Philippine peso coins.

Supervised learning is a machine learning methodology in which the neural network is learning from labeled data. The goal of supervised learning is to predict an outcome from a given new input based on the previous inputted data. The advantage of using supervised learning is it can easily be used in classification and regression. In contrast, its disadvantage is the difficulty in collecting labeled data sets [6].

The Dataset



Figure 1: Sample images from the dataset

The data sets that will be used in this study were obtained from Kaggle Repositories [2][12], a popular platform for computer science and machine learning. The data sets obtained were not directly what the study needed but just a portion of it. Hence, it was necessary to create a labeled data set of our own that would be used in training the neural network. The data set is composed of images of Philippine peso coins starting from twenty-five cents to 20 pesos with a quantity of 100 images each. The images vary in terms of lighting conditions, camera distance, camera angle, background, rotation, and noise.

The obtained data set will require prepossessing along with the created data set before being used in the neural network. Each image will be labeled, resized to 128x128 pixels, and will be categorized in terms of value. The coin images contain both recently developed and past designs of the Philippine peso coins that are still being used in the present. The data set will be divided into three: training, validation, and testing. The distribution of the data set will depend on how well the program works using different methods of training the neural network.

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